Rectangle Mania

AlgoExpert · Very Hard · Graphs

The problem

- Count axis-aligned rectangles whose four corners are all in the point set.
- Each pair forming a strictly up-right diagonal identifies one rectangle exactly once.

Intuition

- Two points forming an up-right diagonal define a rectangle if the other two corners exist.

Brute force vs optimal

- All 4-point combos → $O(n^4)$
- Diagonal pairs + hash set → $O(n^2)$

Approach

- 1. Put points in a set
- 2. For each pair up and to the right
- 3. Check the other two corners
- 4. Count

Worked example

```
[(0, 0), (0, 1), (1, 1), (1, 0), (2, 1), (2, 0), (3, 1), (3, 0)]
```

Points

Worked example

```
pair (a, b) with b up-right of a = a diagonal

check the other two corners in a set
```

Bottom-left & top-right as diagonal

Worked example

```
rectangles: 6
```

Answer: 6

Java solution

```java
public class RectangleMania {
    public static int rectangleMania(int[][] coords) {
        Set<String> set = new HashSet<>();
        for (int[] p : coords) set.add(p[0] + "," + p[1]);
        int count = 0;
        for (int[] a : coords) for (int[] b : coords)
            if (b[0] > a[0] && b[1] > a[1] && set.contains(a[0] + "," + b[1]) && set.contains(b[0] + "," + a[1])) count++;
        return count;
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n)

Edge cases & pitfalls

- Strictly up-right avoids double counting
- Duplicate points

Interview tips & follow-ups

- Contrast with Count Squares

Recap

- Two points forming an up-right diagonal define a rectangle if the other two corners exist.
- Complexity: O(n^2)